Gizem Toplu Tutay

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Education

The University of Texas at Austin **Austin, TX**
PhD in Operations Research and Industrial Engineering *Jan 2021 – Jun 2025 (expected)*
Focus: Stochastic and mixed-integer programming (MIP).
Relevant coursework: Data Science Lab, Linear Programming, Integer Programming, Nonlinear Programming, Decision Analysis, Optimization in Finance, Logistics, and Statistical Modeling.
Relevant Course Project: Carbon Capture and Storage Pipeline Network Optimization. MIP.

Bogazici University **Istanbul, Turkey**
BS in Civil Engineering (High Honors, ranked 3rd) *Aug 2008 – May 2013*
Thesis: Investment decision on wind turbines and energy storage. MATLAB Programming.

Skills & Certifications

Programming: Python (Tensorflow, PyTorch, Keras, Numpy, Pandas, Matplotlib, Seaborn, Scikit-learn), SQL, R, MATLAB

Optimizaton: GAMS, Gurobi, CPLEX

AI/ML/DL: Regression, Classification, Clustering, Deep Learning, Recommender Systems, Reinforcement Learning, Computer Vision, Natural Language Processing (NLP)

Certification: Machine Learning Specialization, Stanford University and DeepLearning.AI

Experience

The University of Texas at Austin **Austin, TX**
Graduate Research Assistant *Jan 2021 – Dec 2025 (expected)*
Stochastic MIP and a Greedy Heuristic for patient evacuation during extreme flooding
CNN for Post-Hurricane Flood Detection
Stochastic MIP to enhance power grid resilience against extreme events by considering equity.
INFORMS and IISE presentations. Publication count: 1 published, 1 submitted, 1 working paper.
Teaching: Led "Engineering Statistics" recitations. Assisted in "Learning Structures" graduate course.

GISEMAI **Ankara, Turkiye**
Founder *Jan 2018 – Dec 2020*
Led e-commerce startup selling in North America. Applied Neural Networks for sales forecasting.

CLK Energy Investment **Istanbul, Turkiye**
Senior Strategic Planning Analyst *Jun 2015 – Nov 2017*
Led team of two. Worked with the departments across the enterprise. Influenced strategic decisions. Applied Time Series Forecasting in R programming for electricity demand prediction.

Independent Projects

Biodiversity Monitoring using CNN and Transfer Learning: pretrained NASNet. F-1 85%.

Anomaly Detection for Credit Card Fraud: Gaussian Distribution. Kaggle Dataset. F1 score 77%.

NLP for Emergency Text Analysis: Topic Modeling with Latent Dirichlet Allocation. High Coherence Score (-2.43).

Publications

- G. Toplu-Tutay, J.J. Hasenbein, E. Kutanoglu (2022). "A Scenario-based Optimization Model for Long-term Healthcare Infrastructure Resilience against Flooding," Proceedings of the IISE Annual Conference & Expo.
- G. Toplu-Tutay, J.J. Hasenbein, E. Kutanoglu. "Impact of Power Outages Depends on Who Loses It: Equity-Informed Grid Resilience Planning via Stochastic Optimization," (Submitted).
- K.Y. Kim, G. Toplu-Tutay, J.J. Hasenbein, E. Kutanoglu. "A Stochastic Optimization Model for Patient Evacuation from Health Care Facilities during Hurricanes," (Working Paper).

Notable Activities

- Public Speaking: Led annual seminars at CLK Energy Investment. Presented on electricity market developments to audience of over 50 people managing 25% of Turkiye's total electricity demand (2015-2017).
- Leadership: Coordinated 4 conferences in Turkiye, Azerbaijan (2), and Slovenia on behalf of International Association for Energy Economics (IAEE), Ohio, USA (2014-2016).
- Scholastic Award: Received student excellence scholarship for academic success (2008-2013).
- Athletic Accomplishment: Silver Medalist in National Synchronized Ice Skating Competitions (2000-2008).